

Analysis of Boolean Functions

from CS6817 and O'Donnell.

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1. Boolean function and the Fourier expansion

Definition 1.1: A *boolean function* is a map $\{0, 1\}^n \rightarrow \{0, 1\}$. We may view the domain $\{0, 1\}^n$ sometimes as $\mathbb{F}_2^n$ and sometimes as subsets of $\mathbb{R}^n$. Alternatively, we may use the space $\{-1, 1\}^n$, known as the *Hamming cube*, as our domain.

Definition 1.2: The *Hamming distance* between $x, y \in \{-1, 1\}^n$ is defined as

$$\Delta(x, y) = \#\{i : x_i \neq y_i\}.$$

1.1. Basic Fourier Analysis

The *Fourier expansion* of a Boolean function is essentially a representation of the function as a real, multilinear polynomial. Intuitively, the existence of such a representation can be justified by noting that any factor of x_i^2 can be replaced simply by 1.

We can compute these just via taking a linear combination of indicator variables (that is, use an idea similar to the construction Lagrange polynomials!). For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$, define the indicator basis $\{\mathbb{1}_a\}_{a \in \{-1, 1\}^n}$ and take the appropriate linear combination.

$$\mathbb{1}_a(x) = \prod_{a_i \in a} \left(\frac{1 + a_i x_i}{2} \right),$$

$$f = \sum_{a \in \{-1, 1\}^n} f(a) \cdot \mathbb{1}_a.$$

Definition 1.1.1: A subset $S \subseteq [n]$ corresponds to a monomial

$$x^S = \prod_{i \in S} x_i.$$

Furthermore,

$\hat{f}(S)$ = coefficient on monomial x^S in the multilinear representation of f .

Proposition 1.1.1: Every function $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ can be uniquely expressed as a multilinear polynomial

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S) x^S.$$

This expression is called the *Fourier expansion* of f , and the real number $\hat{f}(S)$ is called the *Fourier coefficient* of f on S . Collectively, the coefficients are called the *Fourier spectrum* of f .

Definition 1.1.2: We define the encoding $\chi : \mathbb{F}_2 \rightarrow \mathbb{R}$ as follows: for any $b \in \mathbb{F}_2$,

$$\chi(b) = (-1)^b.$$

More explicitly,

$$\chi(0_{\mathbb{F}_2}) = +1, \quad \chi(1_{\mathbb{F}_2}) = -1.$$

Definition 1.1.3: For $S \subseteq [n]$, we define $\chi_S : \mathbb{F}_2^n \rightarrow \mathbb{R}$ by

$$\chi_S(\mathbf{x}) = \prod_{i \in S} \chi(x_i) = (-1)^{\sum_{i \in S} x_i}.$$

Lemma 1.1.1: We have $\chi_S(\mathbf{x} + \mathbf{y}) = \chi_S(\mathbf{x}) \cdot \chi_S(\mathbf{y})$.

The χ_S compute the logical parity of the bits $(x_i)_{i \in S}$. For consistency in notation, we may also define $\chi_S : \{-1, 1\}^n \rightarrow \{-1, 1\}$ by

$$\chi_S(\mathbf{x}) = \prod_{i \in S} x_i.$$

By $\mathbf{x} \sim \{-1, 1\}^n$, we denote that $\mathbf{x}$ is a uniformly chosen random string from $\{-1, 1\}^n$.

Definition 1.1.4: We define an inner product $\langle \cdot, \cdot \rangle$ on pairs of functions $f, g : \{-1, 1\}^n \rightarrow \mathbb{R}$ by

$$\langle f, g \rangle = 2^{-n} \sum_{\mathbf{x} \in \{-1, 1\}^n} f(\mathbf{x})g(\mathbf{x}) = \mathbb{E}_{\mathbf{x} \in \{-1, 1\}^n} [f(\mathbf{x})g(\mathbf{x})].$$

We also have $\|f\|_2 = \sqrt{\langle f, f \rangle}$ and more generally,

$$\|f\|_p = \mathbb{E}[|f(\mathbf{x})|^p]^{\frac{1}{p}}.$$

Unless otherwise specified, probabilities and expectation will always be over a uniformly random $\mathbf{x} \sim \{-1, 1\}^n$.

Lemma 1.1.2: For any $\mathbf{x} \in \{-1, 1\}^n$, we have $\chi_S(\mathbf{x})\chi_T(\mathbf{x}) = \chi_{S \Delta T}(\mathbf{x})$

Lemma 1.1.3: For any $\mathbf{x} \in \{-1, 1\}^n$, we have

$$\mathbb{E}[\chi_S(\mathbf{x})] = \mathbb{E}\left[\prod_{i \in S} x_i\right] = \begin{cases} 1 & \text{if } S = \emptyset, \\ 0 & \text{if } S \neq \emptyset. \end{cases}$$

Proposition 1.1.2: The 2^n parity functions $\chi_S : \{-1, 1\}^n \rightarrow \{-1, 1\}$ form an orthonormal basis for the vector space V of functions $\{-1, 1\}^n \rightarrow \mathbb{R}$

$$\langle \chi_S, \chi_T \rangle = \begin{cases} 1 & \text{if } S = T, \\ 0 & \text{if } S \neq T. \end{cases}$$

Proposition 1.1.3: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ and $S \subseteq [n]$, the Fourier coefficient of f on S is given by

$$\hat{f}(S) = \langle f, \chi_S \rangle = \mathbb{E}[f(\mathbf{x})\chi_S(\mathbf{x})].$$

Proposition 1.1.4 (Parseval's Theorem): For any $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

$$\langle f, f \rangle = \mathbb{E}[f(\mathbf{x})^2] = \sum_{S \subseteq [n]} \hat{f}(S)^2.$$

Proposition 1.1.5 (Plancharel's Theorem): For any $f, g : \{-1, 1\}^n \rightarrow \mathbb{R}$,

$$\langle f, g \rangle = \mathbb{E}[f(\mathbf{x})g(\mathbf{x})] = \sum_{S \subseteq [n]} \hat{f}(S)\hat{g}(S).$$

For Boolean-valued functions, $\langle f, g \rangle$ can be interpreted as a similarity measure between f and g .

Definition 1.1.5: Given $f, g : \{-1, 1\}^n \rightarrow \{-1, 1\}$, we define their *relative Hamming distance* to be

$$\text{dist}(f, g) = \mathbb{P}[f(\mathbf{x}) \neq g(\mathbf{x})],$$

the fraction of inputs on which they disagree.

Proposition 1.1.6: If $f, g : \{-1, 1\}^n \rightarrow \{-1, 1\}$,

$$\langle f, g \rangle = 1 - 2 \text{dist}(f, g).$$

Proof:

$$\langle f, g \rangle = \mathbb{P}[f(\mathbf{x}) = g(\mathbf{x})] - \mathbb{P}[f(\mathbf{x}) \neq g(\mathbf{x})] = 1 - 2 \mathbb{P}[f(\mathbf{x}) \neq g(\mathbf{x})] = 1 - 2 \text{dist}(f, g).$$

□

Definition 1.1.6: The *mean* of $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ is $\mathbb{E}[f]$. If $\mathbb{E}[f] = 0$ then we say that f is *unbiased* or *balanced*. Particularly,

$$\mathbb{E}[f] = \mathbb{P}[f(\mathbf{x}) = 1] - \mathbb{P}[f(\mathbf{x}) \neq 1].$$

Lemma 1.1.4: If $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ then f is balanced iff $\hat{f}(\emptyset) = 0$.

Proof:

$$\mathbb{E}[f] = 0 \Leftrightarrow \langle f, 1 \rangle = 0 \Leftrightarrow \hat{f}(\emptyset) = 0.$$

□

Now, we move on to discussing the variance for Boolean functions.

Proposition 1.1.7: The variance of $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ is

$$\text{Var}[f] = \sum_{S \neq \emptyset} \hat{f}(S)^2.$$

Proof:

$$\mathbf{Var}[f] = \langle f - \mathbb{E}[f], f - \mathbb{E}[f] \rangle = \mathbb{E}[f^2] - \mathbb{E}[f]^2 = \sum_{S \neq \emptyset} \hat{f}(S)^2.$$

□

Proposition 1.1.8: The covariance of $f, g : \{-1, 1\}^n \rightarrow \mathbb{R}$ is

$$\mathbf{Cov}[f, g] = \sum_{S \neq \emptyset} \hat{f}(S) \hat{g}(S).$$

For Boolean-valued functions, the variance has a stronger interpretation. It is roughly proportional to a function's distance from being constant.

Lemma 1.1.5: For $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, we have

$$\mathbf{Var}[f] = 4 \mathbb{P}[f(\mathbf{x}) = 1] \mathbb{P}[f(\mathbf{x}) = -1] \in [0, 1]$$

Proof:

$$\mathbf{Var}[f] = \mathbb{E}[f^2] - \mathbb{E}[f]^2 = 1 - \mathbb{E}[f]^2 = 4 \mathbb{P}[f(\mathbf{x}) = 1] \mathbb{P}[f(\mathbf{x}) = -1] \in [0, 1].$$

□

Lemma 1.1.6: Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$. Then $2\varepsilon \leq \mathbf{Var}[f] \leq 4\varepsilon$ where

$$\varepsilon = \min\{\text{dist}(f, 1), \text{dist}(f, -1)\}.$$

Proof: We have,

$$2\varepsilon \leq \mathbf{Var}[f] = 4 \text{dist}(f, 1) \text{dist}(f, -1) \leq 4\varepsilon.$$

□

Finally, we discuss the Fourier-weight distribution.

Definition 1.1.7: The (*Fourier*) *weight* of $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ on a set S is $\hat{f}(S)^2$.

Definition 1.1.8: Given $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ the *spectral sample* of f , denoted $\mathcal{S}_f$, is the probability distribution on subsets of $[n]$ in which the set S has probability $\hat{f}(S)^2$. We write $S \sim \mathcal{S}_f$ for a draw from this distribution.

Definition 1.1.9: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ and $0 \leq k \leq n$, the (Fourier) weight of f at degree k is

$$W^k[f] = \sum_{\substack{S \subseteq [n] \\ |S| = k}} \hat{f}(S)^2.$$

If $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is Boolean-valued, an equivalent definition is

$$W^k[f] = \mathbb{P}_{S \sim \mathcal{S}_f} [|S| = k].$$

We further define,

$$f^{\leq k} = \sum_{|S| \leq k} \hat{f}(S) \chi_S, \quad f^{=k} = \sum_{|S| = k} \hat{f}(S) \chi_S, \quad f^{>k} = \sum_{|S| > k} \hat{f}(S) \chi_S.$$

Then, $W^{=k} = \|f^{=k}\|_2^2$, $W^{\leq k} = \|f^{\leq k}\|_2^2$ and $W^{>k} = \|f^{>k}\|_2^2$.

1.2. Probability Densities

We have, when possible, been presenting results in the setting of real-valued Boolean functions. An important class of such functions are probability densities.

Definition 1.2.1: A (probability) density function on the Hamming cube $\mathbb{F}_2^n$ is a nonnegative function $\varphi : \mathbb{F}_2^n \rightarrow \mathbb{R}^{\geq 0}$ such that

$$\mathbb{E}[\varphi(\mathbf{x})] = 1.$$

We write $\mathbf{y} \sim \varphi$ to denote that $\mathbf{y}$ is a random string drawn from the distribution defined by

$$\mathbb{P}_{\mathbf{y} \sim \varphi} [\mathbf{y} = \mathbf{z}] = \varphi(\mathbf{z}) \frac{1}{2^n}.$$

Example: If $A \subseteq \mathbb{F}_2^n$ we write $\mathbb{1}_A : \mathbb{F}_2^n \rightarrow \{0, 1\}$ for the 0-1 indicator function of A ,

$$\mathbb{1}_A(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{x} \in A, \\ 0 & \text{if } \mathbf{x} \notin A. \end{cases}$$

The associated relative density function is obtained by normalizing this,

$$\varphi_A = \frac{1}{\mathbb{E}[\mathbb{1}_A]} \cdot \mathbb{1}_A.$$

A classical reason for using the Fourier basis is that it interacts particularly nicely with *convolution*.

Definition 1.2.2: Let $f, g : \mathbb{F}_2^n \rightarrow \mathbb{R}$. Their *convolution* is the function $f \star g : \mathbb{F}_2^n \rightarrow \mathbb{R}$ defined by

$$(f \star g)(\mathbf{x}) = \mathbb{E}_{\mathbf{y} \sim \mathbb{F}_2^n} [f(\mathbf{y})g(\mathbf{x} - \mathbf{y})] = \mathbb{E}[f(\mathbf{x} - \mathbf{y})g(\mathbf{y})]$$

Over $\mathbb{F}_2$, subtraction and addition are equivalent. So,

$$(f \star g)(\mathbf{x}) = \mathbb{E}_{\mathbf{y} \sim \mathbb{F}_2^n} [f(\mathbf{y})g(\mathbf{x} + \mathbf{y})] = \mathbb{E}[f(\mathbf{x} + \mathbf{y})g(\mathbf{y})].$$

Lemma 1.2.1: If φ is a density function on $\mathbb{F}_2^n$ and $g : \mathbb{F}_2^n \rightarrow \mathbb{R}$ then

$$(\varphi \star g)(\mathbf{x}) = \mathbb{E}_{\mathbf{y} \sim \varphi} [g(\mathbf{x} - \mathbf{y})] = \mathbb{E}_{\mathbf{y} \sim \varphi} [g(\mathbf{x} + \mathbf{y})].$$

In particular, $\mathbb{E}_{\mathbf{y} \sim \varphi} [g(\mathbf{y})] = \varphi \star g(0)$.

Proposition 1.2.1: If φ, ψ are probability distributions then so is $\varphi \star \psi$. It represents the distribution on $\mathbf{x} \in \mathbb{F}_2^n$ given by choosing $\mathbf{y} \sim \varphi$ and $\mathbf{z} \sim \psi$ independently and setting $\mathbf{x} = \mathbf{y} + \mathbf{z}$.

Proposition 1.2.2: Let $f, g : \mathbb{F}_2^n \rightarrow \mathbb{R}$. Then for all $S \subseteq [n]$,

$$\widehat{f \star g}(S) = \hat{f}(S) \hat{g}(S).$$

2. Basic concepts and social choice

In this section, we explore connections between social choice and Fourier analysis. We will see that not only is the latter useful for the former but the former is a treasure trove of intuition for the latter.

Definition 2.1: For n odd, the *majority* function $\text{Maj}_n : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is defined by

$$\text{Maj}_n(\mathbf{x}) = \text{sgn}(x_1 + \dots + x_n).$$

The Boolean AND and OR correspond to voting rules in which a certain candidate is always elected unless all voters are unanimously opposed.

Definition 2.2: The function $\text{AND}_n : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is defined by

$$\text{AND}_n(\mathbf{x}) = \begin{cases} -1 & \text{if } \mathbf{x} = (-1, -1, \dots, -1), \\ +1 & \text{otherwise.} \end{cases}$$

The function $\text{OR}_n : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is defined by

$$\text{OR}_n(\mathbf{x}) = \begin{cases} +1 & \text{if } \mathbf{x} = (+1, +1, \dots), \\ -1 & \text{otherwise.} \end{cases}$$

Dictators play an important role in both social choice and Fourier analysis.

Definition 2.3: The *i-th dictator* function $\chi_i : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is defined by $\chi_i(\mathbf{x}) = x_i$

We further generalize the dictator to function that only depend on a constant number of coordinates.

Definition 2.4: A function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is called a *k-junta* for $k \in \mathbb{N}$ if it depends on at most k of its input coordinates. That is,

$$f(\mathbf{x}) = g(x_{i_1}, x_{i_2}, \dots, x_{i_k})$$

for some $g : \{-1, 1\}^k \rightarrow \{-1, 1\}$ and $i_1, \dots, i_k \in [n]$.

The EU's Council of Ministers adopts decisions based on a weighted majority voting rule.

Definition 2.5: A function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is called a *weighted majority* or *(linear) threshold function* if it is expressible as

$$f(x) = \text{sgn}(a_0 + a_1x_1 + \dots + a_nx_n)$$

for some $a_0, a_1, \dots, a_n \in \mathbb{R}$.

The President of the US is elected via a sort of “two-level majority.” This motivates the following definition,

Definition 2.6: The *depth-d recursive majority of n* function, denoted $\text{Maj}_n^{\otimes d}$ is the boolean function on n^d bits defined inductively as follows:

$$\begin{aligned} \text{Maj}_n^{\otimes 1} &= \text{Maj}_n \\ \text{Maj}_n^{\otimes d+1}(x^1, \dots, x^n) &= \text{Maj}_n(\text{Maj}_n^{\otimes d}(x^1), \dots, \text{Maj}_n^{\otimes d}(x^n)), \quad \text{for } x^i \in \{-1, 1\}^{n^d}. \end{aligned}$$

Tribes_n is a voting rule that is quite important in the Fourier context but perhaps less so in real life. Here, the voters are divided into “tribes” of equal size and the outcome is true if and only if at least one tribe is unanimously in favor of true (a sort of CNF-SAT problem).

Definition 2.7: The *tribes* function of width w and size s , $\text{Tribes}_{s,w} : \{-1, 1\}^{sw} \rightarrow \{-1, 1\}$ is defined by

$$\text{Tribes}_{s,w}(x^1, \dots, x^s) = \text{OR}_s(\text{AND}_w(x^1), \dots, \text{AND}_w(x^s)),$$

for $x^i \in \{-1, 1\}^w$.

In practice, it is desirable for voting rules to have particular properties that we believe a desirable voting system should intuitively exhibit.

Definition 2.8: We say that a function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is

- a. *monotone* if $f(x) \leq f(y)$ whenever $x \leq y$ coordinate-wise;
- b. *odd* if $f(-x) = -f(x)$;
- c. *unanimous* if $f(1, \dots, 1) = 1$ and $f(-1, \dots, -1) = -1$;
- d. *symmetric* if $f(x^\pi) = f(x)$ for all permutations $\pi \in S_n$.

Maj_n satisfies all these properties. The dictators and recursive majority satisfy the first three properties. The AND and OR are monotone, unanimous and symmetric, but not odd. The tribes functions are monotone and unanimous; they satisfy a weaker transitive property.

Definition 2.9: A function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is *transitive-symmetric* if for all $i, i' \in [n]$ there exists a permutation $\pi \in S_n$ taking i to i' such that $f(x^\pi) = f(x)$ for all $x \in \{-1, 1\}^n$.

2.1. Influence and Derivatives

Given a voting rule $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ we are interested in investigating the *influence* of the i th voter.

Definition 2.1.1: For $x \in \{-1, 1\}^n$, we defined $x^{\oplus i} = (x_1, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_n)$.

Definition 2.1.2: We say that coordinate $i \in [n]$ is *pivotal* for $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ on an input x if $f(x) \neq f(x^{\oplus i})$.

Definition 2.1.3: The *influence* of coordinate i on $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, is defined to be the probability that i is pivotal for a random input

$$\mathbf{Inf}_i[f] = \mathbb{P}[f(\mathbf{x}) \neq f(\mathbf{x}^{\oplus i})].$$

Definition 2.1.4: We say that coordinant $i \in [n]$ is *relevant* for $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ if and only if $\mathbf{Inf}_i[f] > 0$.

We can also interpret this in terms of the geometry of the Hamming cube. We say that $(\mathbf{x}, \mathbf{y})$ is a dimension- i edge if $\mathbf{y} = \mathbf{x}^{\oplus i}$; it is a boundary edge if $f(\mathbf{x}) \neq f(\mathbf{y})$. The influence $\mathbf{Inf}_i[f]$ equals the fraction of dimension- i edges which are boundary edges.

A more analytic approach to influence is obtained by defining the discrete derivative.

Definition 2.1.5: For $\mathbf{x} \in \{-1, 1\}^n$ and $b \in \{-1, 1\}$, we define $\mathbf{x}^{i \rightarrow b} = (x_1, \dots, x_{i-1}, b, x_{i+1}, \dots, x_n)$.

Definition 2.1.6: The i th (discrete) derivative D_i maps the function $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ to the function $D_i f : \{-1, 1\}^n \rightarrow \mathbb{R}$ defined by

$$D_i f(\mathbf{x}) = \frac{f(\mathbf{x}^{i \rightarrow 1}) - f(\mathbf{x}^{i \rightarrow -1})}{2}$$

Proposition 2.1.1: If $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, $D_i f(\mathbf{x})^2$ is the 0-1 indicator for whether i is pivotal for $\mathbf{x}$. That is,

$$\mathbf{Inf}_i[f] = \mathbb{E}[D_i f(\mathbf{x})^2] = \|D_i f\|_2^2.$$

Proof: This follows from noting that

$$D_i f(\mathbf{x}) = \begin{cases} 0 & \text{if coordinate } i \text{ is not pivotal for } \mathbf{x}, \\ \pm 1 & \text{if coordinate } i \text{ is pivotal for } \mathbf{x}. \end{cases}$$

□

The discrete derivative is analogous to partial derivatives.

Proposition 2.1.2: Let $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ have the multilinear expansion $f(\mathbf{x}) = \sum_{S \subseteq [n]} \hat{f}(S) x^S$. Then,

$$D_i f(\mathbf{x}) = \sum_{\substack{S \subseteq [n] \\ S \ni i}} \hat{f}(S) x^{S \setminus \{i\}}.$$

This gives us a Fourier formula for influence.

Proposition 2.1.3: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ and $i \in [n]$,

$$\mathbf{Inf}_i[f] = \sum_{S \ni i} \hat{f}(S)^2.$$

For monotone functions, we have a simpler result.

Proposition 2.1.4: If $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is monotone, then $\mathbf{Inf}_i[f] = \hat{f}(i)$.

Proof: By monotonicity, $D_i f$ is $+1$ whenever i is pivotal. Thus, $D_i f$ itself is a 0-1 indicator for i being pivotal. Thus,

$$\mathbf{Inf}_i[f] = \mathbb{E}[D_i f(\mathbf{x})] = \widehat{D_i f}(\emptyset) = \hat{f}(i).$$

□

Here is an interesting consequence for 2-candidate voting rules that are monotone and transitive-symmetric.

Proposition 2.1.5: Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ be transitive-symmetric and monotone. Then $\mathbf{Inf}_i[f] \leq \frac{1}{\sqrt{n}}$ for all $i \in [n]$.

Proof: Transitive-symmetry of f implies that $\hat{f}(i) = \hat{f}(i')$ for all $i, i' \in [n]$. By monotonicity, $\mathbf{Inf}_i[f] = \hat{f}(i) = \hat{f}(1)$ for all $i \in [n]$. By Parseval,

$$\begin{aligned} 1 &= \sum_S \hat{f}(S)^2 \geq \sum_{i=1}^n \hat{f}(i)^2 = n \hat{f}(1)^2 \\ &\Rightarrow \hat{f}(1) \leq \frac{1}{\sqrt{n}}. \end{aligned}$$

□

The derivative operator doesn't generalize well to domains other than $\{-1, 1\}^n$. Here are some ideas that do generalize better.

Definition 2.1.7: The i -th expectation operator $\mathbb{E}_i$ is the linear operator on functions $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ defined by

$$\mathbb{E}_i[f(\mathbf{x})] = \mathbb{E}_y[f(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n)].$$

Whereas $D_i f$ isolates the part of f depending on the i -th coordinate, $\mathbb{E}_i f$ isolates the part of f not depending on the i -th coordinate.

Proposition 2.1.6: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

- $\mathbb{E}_i f(\mathbf{x}) = \frac{f(\mathbf{x}^{i \rightarrow 1}) + f(\mathbf{x}^{i \rightarrow -1})}{2},$
- $\mathbb{E}_i f(\mathbf{x}) = \sum_{S \not\ni i} \hat{f}(S) x^S,$
- $f(\mathbf{x}) = x_i D_i f(\mathbf{x}) + \mathbb{E}_i f(\mathbf{x}).$

We also define a Laplacian operator.

Definition 2.1.8: The i -th coordinate Laplacian operator $\mathcal{L}_i$ is defined by

$$\mathcal{L}_i f = f - \mathbb{E}_i f.$$

The following properties are useful.

Proposition 2.1.7: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

- a. $\mathcal{L}_i f(\mathbf{x}) = \frac{f(\mathbf{x}) - f(\mathbf{x}^\oplus)}{2}$,
- b. $\mathcal{L}_i f(\mathbf{x}) = x_i D_i f(\mathbf{x}) = \sum_{S \ni i} \hat{f}(S) x^S$,
- c. $\langle f, \mathcal{L}_i f \rangle = \langle \mathcal{L}_i f, \mathcal{L}_i f \rangle = \mathbf{Inf}_i[f]$.

2.2. Total Influence

Definition 2.2.1: The *total influence* of $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ is defined to be

$$\mathbf{Inf}[f] = \sum_{i=1}^n \mathbf{Inf}_i[f].$$

The following proves a useful interpretation for total influence.

Definition 2.2.2: For $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, the *sensitivity* of f at $\mathbf{x}$, $\text{sens}_f(\mathbf{x})$, is defined to be the number of pivotal coordinates for f on input $\mathbf{x}$.

Proposition 2.2.1: For $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$,

$$\mathbf{Inf}[f] = \mathbb{E}_{\mathbf{x}}[\text{sens}_f(\mathbf{x})].$$

The total influence for Boolean-valued functions is related to the size of a function's edge boundary.

Lemma 2.2.1: The fraction of edges in the Hamming cube $\{-1, 1\}^n$ which are boundary edges for $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is equal to $\frac{1}{n} \mathbf{Inf}[f]$.

The sum of the degree-1 coefficients has a natural interpretation in social choice.

Proposition 2.2.2: Let $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ be a voting rule for a 2-candidate election. Given votes $\mathbf{x} = (x_1, \dots, x_n)$, let w be the number of votes that agree with the outcome of the election $f(\mathbf{x})$. Then,

$$\mathbb{E}[w] = \frac{n}{2} + \frac{1}{2} \sum_{i=1}^n \hat{f}(i).$$

For monotone functions, we have already seen the degree-1 coefficients under a different name.

Proposition 2.2.3: If $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ is monotone, then

$$\mathbf{Inf}[f] = \sum_{i=1}^n \hat{f}(i).$$

In fact, from the perspective of maximizing the number of votes that agree with the outcome, the majority rule is optimal (at least for odd n).

Proposition 2.2.4: The unique maximizers of $\sum_{i=1}^n \hat{f}(i)$ among all $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ are the majority functions. In particular,

$$\mathbf{Inf}[f] \leq \mathbf{Inf}[\text{Maj}_n] = \sqrt{\frac{2n}{\pi}} + O(n^{-\frac{1}{2}}).$$

We now define the discrete gradient operator to give another expression for the total influence.

Definition 2.2.3: The *(discrete) gradient operator* ∇ maps the function $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ to the function $\nabla f : \{-1, 1\}^n \rightarrow \mathbb{R}^n$ defined by

$$\nabla f = (D_1 f(\mathbf{x}), \dots, D_n f(\mathbf{x})).$$

Proposition 2.2.5: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

$$\mathbf{Inf}[f] = \mathbb{E}_{\mathbf{x}} [\|\nabla f(\mathbf{x})\|_2^2].$$

Finally, the Laplacian also provides a perspective on the total influence.

Definition 2.2.4: The *Laplacian operator* $\mathcal{L}$ is the linear operator on functions $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ defined by

$$\mathcal{L} = \sum_{i=1}^n \mathcal{L}_i.$$

Proposition 2.2.6: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

- a. $\mathcal{L}f(\mathbf{x}) = \frac{n}{2} (f(\mathbf{x}) - \text{avg}_{i \in [n]} \{f(x^{\oplus i})\})$,
- b. $\mathcal{L}f(\mathbf{x}) = f(\mathbf{x}) \text{sens}_{f(\mathbf{x})}$ if $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$,
- c. $\mathcal{L}f = \sum_{S \subseteq [n]} |S| \hat{f}(S) \chi_S$,
- d. $\langle f, \mathcal{L}f \rangle = \mathbf{Inf}[f]$.

Finally, we can express the total influence directly in terms of the Fourier spectrum and the spectral sample (that is, in some sense, the total influence also measures the average degree of the Fourier weights).

Proposition 2.2.7: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$,

$$\mathbf{Inf}[f] = \sum_{S \subseteq [n]} |S| \cdot \hat{f}(S)^2 = \sum_{k=0}^n k \cdot W^k[f].$$

For $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, we can express this using the spectral sample,

$$\mathbf{Inf}[f] = \mathbb{E}_{S \sim \mathcal{S}_f} [|S|].$$

2.3. Noise Stability

Consider a situation where there is some chance that a vote is misrecorded. We formalize this in terms of noise stability.

Definition 2.3.1: Let $\rho \in [-1, 1]$. We write $\mathbf{y} \sim N_\rho(\mathbf{x})$ to denote the random string $\mathbf{y}$ is drawn as follows

$$y_i = \begin{cases} x_i & \text{with probability } \frac{1}{2} + \frac{1}{2}\rho, \\ -x_i & \text{with probability } \frac{1}{2} - \frac{1}{2}\rho. \end{cases}$$

We say that $\mathbf{y}$ is ρ -correlated to $\mathbf{x}$.

Definition 2.3.2: If $\mathbf{x} \sim \{-1, 1\}^n$ is drawn uniformly at random and then $\mathbf{y} \sim N_\rho(\mathbf{x})$, we say that $(\mathbf{x}, \mathbf{y})$ is a ρ -correlated pair of random strings. That is, for each $i \in [n]$, the pair of random bits (x_i, y_i) satisfies

$$\begin{aligned} \mathbb{E}[x_i] &= \mathbb{E}[y_i] = 0 \\ \mathbb{E}[x_i y_i] &= \rho. \end{aligned}$$

Definition 2.3.3: For $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ and $\rho \in [-1, 1]$, the *noise stability* of f at ρ is

$$\text{Stab}_\rho[f] = \mathbb{E}_{\substack{(\mathbf{x}, \mathbf{y}) \\ \rho\text{-correlated}}} [f(\mathbf{x})f(\mathbf{y})].$$

If $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ we have

$$\begin{aligned} \text{Stab}_\rho[f] &= \mathbb{P}_{\substack{(\mathbf{x}, \mathbf{y}) \\ \rho\text{-correlated}}} [f(\mathbf{x}) = f(\mathbf{y})] - \mathbb{P}_{\substack{(\mathbf{x}, \mathbf{y}) \\ \rho\text{-correlated}}} [f(\mathbf{x}) \neq f(\mathbf{y})] \\ &= 2 \mathbb{P}_{\substack{(\mathbf{x}, \mathbf{y}) \\ \rho\text{-correlated}}} [f(\mathbf{x}) = f(\mathbf{y})] - 1. \end{aligned}$$

When ρ is closed to 1, it's sometimes more natural to ask about the probability that reversing a small fraction of the votes reverses the outcome of the election.

Definition 2.3.4: For $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$ and $\delta \in [0, 1]$ we write $\text{NS}_\delta[f]$ for *noise sensitivity* of f at δ , defined to be the probability that $f(\mathbf{x}) \neq f(\mathbf{y})$ when $\mathbf{x} \sim \{-1, 1\}^n$ is uniformly random and $\mathbf{y}$ is formed from $\mathbf{x}$ by reversing each bit with probability δ . In other words,

$$\text{NS}_\delta[f] = \frac{1}{2} - \frac{1}{2} \text{Stab}_{1-2\delta}[f].$$

3. Spectral structure and learning
4. DNF formulas and small-depth circuits
5. Generalized domains
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7. Pseudorandomness and $\mathbb{F}_2$ -polynomials